

Ryman J. Mahnke

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EDUCATION

University of California Los Angeles, Samueli School of Engineering
Bachelor of Science, Mechanical Engineering

Expected June 2029
GPA 3.974

EXPERIENCE

Responsible Engineer, Bruin Formula Racing

Sep. 2025 - Present

- **Mechanical CAD Design:** Engineered high-stress motor and differential mounts using SolidWorks to adhere to strict chassis constraints and dynamic powertrain loads.
- **FEA & Topology Optimization:** Executed structural finite element analysis (FEA) and topology optimization in Ansys, reducing assembly weight while preserving torsional rigidity under peak torque conditions.
- **Precision CNC Machining:** Programmed 3-axis CNC toolpaths and G-code in Fusion 360; setup and milled custom components maintaining tight tolerances.
- **Cross-Functional Integration:** Collaborated across chassis, suspension, and driver interface subteams to resolve spatial interferences and streamline system integration.
- **Component Development:** Designed and fabricated custom CV boots and pedal box assemblies to enhance driver control and driveline reliability.
- **Other Components:** Designed and manufactured custom CV boots and pedal box components as well.

Window Cleaner & Salesperson, Pine Window Cleaning

Jun. 2026 - Aug. 2026

- **Client Relations:** Conducted Sales outreach, negotiated service packages, and expanded local customer base.
- **Service Management:** Managed end-to-end sales from initial contact to cleaning service and ensured high customer retention.
- **Revenue:** Generated \$10,000 in revenue each month.

Engineer Intern, Carollo Engineers

May 2024

- **Schematic & Layout Review:** Analyzed complex hydraulic Piping and Instrumentation Diagrams (P&IDs) for a regional municipal water facility following industry drafting standards.
- **Design & Compliance Reviews:** Participated in multi-disciplinary engineering reviews with civil, mechanical, and environmental leads to track regulatory compliance and safety codes.
- **Water Treatment Pilot Analysis:** Assessed active pilot plant operations dedicated to advanced PFAS remediation methodologies and operational workflow testing

Engineer Job Shadow, MKS Instruments

Jan. 2024 - May 2024

- **Hardware Manufacturing:** Evaluated end-to-end component lifecycles, tight-tolerance machining, and quality assurance workflows for advanced semiconductor equipment.
- **CAD Prototyping & Business Operations:** Designed a custom training component using CAD and analyzed cross-functional dependencies between hardware engineering and product cost estimation.

HONORS AND AWARDS

Dean's Honor List, UCLA
General Engineering Scholarship Recipient, UCLA

2025-2026
Received Sep. 2025

TECHNICAL SKILLS & INTERESTS

- **Technical Skills & Tools:** SolidWorks | Ansys (FEA & Topology Optimization) | Fusion 360 | 3-Axis CNC Milling | CAM Toolpathing | Precision Machining | Python
- **Professional & Interpersonal Strengths:** Cross-Functional Team Collaboration | Technical Communication | Complex Problem-Solving | Conflict Resolution | Adaptability
- **Interests:** Automotive & Electric Vehicle Development, Renewable Energy Grid Infrastructure, Water Resource Management, Environmental Remediation.